SOMAIYA VIDYAVIHAR UNIVERSITY

21·05·2025 (M)

Semester: Jan 2025– May 2025		
Maximum Marks: 50 Examination: End-Semester Examination (Re-ESE) Duration: 2 Hrs.		
Programme code: 01 Programme: B. Tech. (Computer Engineering)	Class: LY	Semester: VIII (SVU-2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMP/~~ETRX/EXTC/IT/MECH/RAI/AIDS/CCE/EXCP~~	
Course Code: 116U01E853	Name of the Course: Internet of Everything	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Compare IoE and M2M.	05
ii)	Explain SDN architecture with reference to IoT.	05
iii)	Short note on IoT applications to enhance human lifestyle.	05
Q.2	Attempt any **one**	10
i)	What are the challenges faced while using IoT in Business Process?	
ii)	What is Edge Analytics? Explain how edge analytics would be useful in IoT application.	
Q.3	Attempt any **one**	10
i)	What are the various levels of IoT implementation? Give suitable example of each.	
ii)	Explain any four types of IoT Communication models.	
Q.4	Attempt the following	10
	a) Explain the use of following functions in Arduino Program i) pinMode() ii) setup() iii) analogWrite()	05
	b) Explain the use of sensors and actuators in IoT. Give examples of few analogue and digital sensors and actuators	05
Q.5	Attempt the following	10
	A farmer has approached you to deliver a system that will optimise usage of water, fertiliser and pesticides. The system should allow display of parameters such as humidity, temperature as well as level of fertiliser. Explain in detail the following steps used in application design: Purpose and Requirements, Process model specification, Functional View Specification, Operation View Specification, and Application development.	